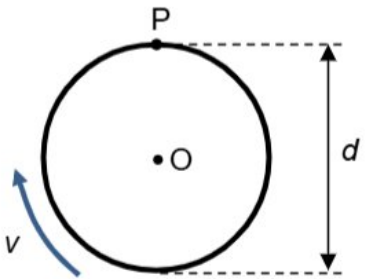
28 An aluminium disc of diameter d rotates about its centre. A point on the rim moves at a constant speed v. It is placed in a uniform magnetic field B perpendicular to its surface. A steady electromotive force (e.m.f.) E is generated between the centre O and the rim at P.



What is the expression for the e.m.f E generated?

A $\frac{Bdv}{4}$

B $\frac{Bdv}{2}$

C Bdv

D Bv